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GRANT NARRATIVE

SLCGP Grant Application Narrative: Cedar Ridge Regional Water Authority

DOCUMENT	State and Local Cybersecurity Grant Program Application Narrative
DATE	April 22, 2026
PREPARED BY	Sentinel OT (on behalf of CRRWA)
CLASSIFICATION	Confidential
REFERENCE	Report ID: pbcwud_cedar-ridge_sample_2026-04-15

Applicant Information

FIELD	DETAIL
Applicant	Cedar Ridge Regional Water Authority (CRRWA)
Type	Municipal water utility, community water system
Population served	Approximately 14,200
AWIA tier	Small system (serving 3,301 to 49,999)
State	[State]
SAA	[State Administrative Agency]
NAICS	221310 (Water Supply and Irrigation Systems)

Executive Summary

Cedar Ridge Regional Water Authority operates a surface water treatment plant serving approximately 14,200 residents across the Cedar Ridge municipal area. In April 2026, a passive intelligence assessment identified two Rockwell Automation CompactLogix programmable logic controllers at the Cedar Ridge Water Treatment Plant directly exposed to the public internet. The assessment confirmed three Critical vulnerabilities (CVSS 9.8), all remotely exploitable without authentication. One of these vulnerabilities, CVE-2021-22681, is listed in the CISA Known Exploited Vulnerabilities (KEV) catalog with a federal remediation deadline that has already passed.

CRRWA is requesting $86,580 in SLCGP funding to conduct a full cybersecurity assessment and remediation of its operational technology infrastructure. The project will eliminate internet exposure of industrial control systems, remediate all confirmed Critical vulnerabilities, establish a defensible network

architecture, and produce compliance-grade documentation meeting AWIA, IEC 62443, and NIST SP 800-82 Rev 3 requirements.

The total project cost is $144,300 ($132,300 in professional services and $12,000 in hardware). CRRWA will provide the required 40% non-federal cost share of $57,720 from its capital improvement fund.

Problem Statement

Current Threat Landscape

Water and wastewater systems are designated critical infrastructure under Presidential Policy Directive 21 (PPD-21) and face an escalating threat environment. CISA, the FBI, and the EPA have issued multiple joint advisories since 2021 warning of nation-state and criminal targeting of water utility SCADA systems, including advisories addressing Iranian-affiliated actors (AA23-335A), Chinese state-sponsored Volt Typhoon campaigns (AA24-038A), and Russian hacktivists targeting HMI systems at water facilities (AA24-010A).

Small and medium water utilities face disproportionate risk due to limited cybersecurity staffing, legacy industrial control systems, and reliance on remote access configurations that were not designed with current threats in mind.

Cedar Ridge Specific Findings

A passive intelligence assessment conducted in April 2026 identified the following confirmed exposures at the Cedar Ridge Water Treatment Plant:

Critical Findings (CVSS 9.8)

CVE	DESCRIPTION	CISA KEV
CVE-2021-22681	Rockwell Automation authentication bypass. Allows an unauthenticated attacker to connect to a CompactLogix controller and read, modify, or replace the running PLC program over CIP (port 44818).	Yes. Federal deadline passed.

CVE	DESCRIPTION	CISA KEV
CVE-2022-1161	Rockwell Automation controller program modification. Exploits the lack of runtime integrity verification to silently alter controller logic without detection by the engineering workstation.	No
CVE-2019-10952	Rockwell Automation web server buffer overflow. A crafted HTTP request to the controller's embedded web server can cause a denial-of-service condition or potentially allow remote code execution.	No

Informational Findings

- XetaWave radio management interface exposed on port 602/tcp
- Open DNS resolver on port 53/tcp
- Default credentials and insecure services (FTP, Telnet) on the cellular gateway

These controllers manage chemical dosing, filtration backwash sequencing, and clearwell level control. Unauthorized modification of these processes could result in public health consequences, regulatory violations, or service disruption affecting 14,200 residents.

Project Description

This project consists of five implementation phases executed over approximately 75 calendar days, followed by continuous monitoring. The project scope and statement of work (referenced as a companion document) provides detailed task-level descriptions for each phase.

Phase 1: Immediate Compensating Controls (Days 1 to 3)

Deploy access control lists and firewall rules on the existing Microhard cellular gateway to block all unauthorized inbound traffic to industrial control system ports. Rotate all credentials and disable insecure services. No plant downtime required.

This phase addresses the most urgent risk by eliminating direct internet accessibility of the PLCs within 72 hours of contract execution.

Phase 2: Comprehensive On-Site Assessment (Days 4 to 14)

Conduct an on-site assessment of the complete OT network at the Cedar Ridge Water Treatment Plant. Validate all passive intelligence findings against physical infrastructure, map the full network architecture, perform controller integrity baselines, and conduct an AWIA-aligned risk and resilience assessment.

Phase 3: Network Architecture Remediation (Days 15 to 45)

Replace the current NAT/port-forwarding remote access model with a VPN-only architecture using certificate-based authentication. Deploy an industrial firewall between the cellular gateway and the PLC subnet. Implement network segmentation per IEC 62443-3-3 zone and conduit requirements.

Phase 4: Controller Remediation (Days 30 to 60)

Upgrade PLC firmware to the latest supported version. Enable CIP Security authentication and integrity features if supported by the hardware platform. If the 1769-L18ER platform does not support CIP Security, document formal risk acceptance with compensating controls and include controller replacement in the FY2027 capital budget.

Phase 5: Documentation and Compliance (Days 45 to 75)

Produce a complete documentation package including an AWIA-compliant Risk and Resilience Assessment, Emergency Response Plan update, standards compliance mapping (IEC 62443, NIST SP 800-82 Rev 3, CISA CPG 2.0), and an executive summary for the CRRWA Board of Directors.

Phase 6: Ongoing Monitoring (Month 3 onward)

Continuous passive monitoring of all CRRWA internet-facing assets, vulnerability alerting for deployed Rockwell Automation products, and quarterly posture reports. This phase is funded separately as an annual subscription and is not included in the SLCGP funding request.

Alignment to SLCGP Eligible Activities

SLCGP OBJECTIVE	PROJECT PHASE	DESCRIPTION
Objective 1: Governance and Planning	Phase 5	AWIA-compliant Risk and Resilience Assessment, Emergency Response Plan update, standards compliance mapping
Objective 2: Assessment and Evaluation	Phases 1 and 2	Passive intelligence validation, on-site comprehensive assessment, controller integrity baselines, vulnerability register
Objective 3: Mitigation	Phases 3 and 4	Network architecture remediation, industrial firewall deployment, VPN implementation, controller firmware upgrades, CIP Security configuration
Objective 4: Workforce Development	Phase 5	Closeout briefing and training for plant operations staff on new access procedures, change management, and incident response

Alignment to NIST Cybersecurity Framework

CSF FUNCTION	PROJECT ACTIVITY
Identify (ID)	Asset inventory, network architecture mapping, vulnerability register, risk and resilience assessment
Protect (PR)	Network segmentation, VPN deployment, multi-factor authentication, credential rotation, industrial firewall, firmware upgrades
Detect (DE)	Controller change detection, continuous passive monitoring, quarterly posture reporting
Respond (RS)	Emergency Response Plan update, incident response capability assessment
Recover (RC)	Controller golden image baselines, backup validation, recovery procedure documentation

Alignment to CISA Cross-Sector Cybersecurity Performance Goals (CPGs)

CPG	PROJECT ACTIVITY
1.A: Mitigate Known Vulnerabilities	Remediation of CVE-2021-22681 (CISA KEV), CVE-2022-1161, CVE-2019-10952
1.E: Default Credentials	Rotation of all default credentials on controllers and network equipment
2.A: Multi-Factor Authentication	MFA requirement for all remote access via secure jump host

CPG	PROJECT ACTIVITY
2.F: Network Segmentation	Zone-based segmentation between PLC, HMI/SCADA, and maintenance networks
2.G: Email Security	Not applicable to this OT-focused engagement
5.A: Asset Inventory	Complete validated asset inventory of OT infrastructure

Budget Justification

Professional Services

PHASE	DESCRIPTION	COST	JUSTIFICATION
Phase 1	Compensating Controls	$4,800	16 hours at $300/hr. Remote configuration of access control lists, credential rotation, service hardening.
Phase 2	On-Site Assessment	$52,000	5 days on-site (2 assessors), plus pre-assessment planning and post-assessment analysis. Includes travel, per diem, and specialized OT assessment tooling. Rate consistent with ICS/OT specialist market rates ($200 to $350/hr).
Phase 3	Network Remediation	$35,000	Industrial firewall deployment, VPN architecture implementation, network segmentation. Requires two scheduled maintenance windows with on-site presence.

PHASE	DESCRIPTION	COST	JUSTIFICATION
Phase 4	Controller Remediation	$22,000	Firmware upgrade coordination with Rockwell Automation, CIP Security configuration, controller integrity baselines. Sequential controller upgrades during maintenance windows.
Phase 5	Documentation	$18,500	AWIA RRA, ERP update, standards compliance mapping, executive summary, closeout briefing. Compliance documentation requires specialized knowledge of AWIA Section 2013, IEC 62443, and NIST SP 800-82.
	Professional Services Total	**$132,300**	

Hardware

ITEM	ESTIMATED COST	JUSTIFICATION
Industrial firewall (e.g., Fortinet FortiGate Rugged 60F or Cisco ISA 3000)	$5,500	Required for IEC 62443-compliant zone segmentation between cellular gateway and PLC subnet. Pricing based on published list prices ($2,770 to $5,750 depending on model selected).
VPN appliance and jump host infrastructure	$4,000	Secure remote access infrastructure replacing current NAT/port-forward model. Includes certificate-based VPN termination and MFA-enabled jump host.

ITEM	ESTIMATED COST	JUSTIFICATION
Cabling, rack hardware, and installation materials	$2,500	Physical installation materials for network architecture changes.
Hardware Total	**$12,000**	

Total Project Budget

CATEGORY	AMOUNT
Professional services	$132,300
Hardware	$12,000
Total project cost	**$144,300**
Federal share (60%)	$86,580
Non-federal match (40%)	$57,720

CRRWA will provide the 40% non-federal cost share from its capital improvement fund. No in-kind match is included in this request.

Project Timeline

PHASE	START	END	DURATION
Phase 1: Compensating Controls	Day 1	Day 3	3 days
Phase 2: On-Site Assessment	Day 4	Day 14	11 days

PHASE	START	END	DURATION
Phase 3: Network Remediation	Day 15	Day 45	31 days
Phase 4: Controller Remediation	Day 30	Day 60	31 days
Phase 5: Documentation	Day 45	Day 75	31 days

Phases 3 and 4 overlap. Total project duration: approximately 75 calendar days from contract execution.

The project can begin within 30 days of grant award. Phase 1 (compensating controls) will be initiated within 72 hours of contract execution to address the immediate risk from CISA KEV-listed vulnerabilities.

Performance Metrics

METRIC	BASELINE (CURRENT)	TARGET (POST-PROJECT)	MEASUREMENT METHOD
Internet-exposed ICS devices	2 PLCs on public internet	0 devices exposed	Passive scan validation
Unpatched Critical CVEs	3 confirmed (CVSS 9.8)	0 unpatched Critical CVEs	Firmware version audit
CISA KEV compliance	1 overdue KEV (CVE-2021-22681)	0 overdue KEVs	KEV catalog cross-reference
Network segmentation	No segmentation (flat network)	Zone-based per IEC 62443	Firewall rule audit
Multi-factor authentication	Not implemented	Required for all remote access	Access configuration audit
AWIA compliance documentation	Not current	Complete and current RRA and ERP	Document review

METRIC	BASELINE (CURRENT)	TARGET (POST-PROJECT)	MEASUREMENT METHOD
Insecure services exposed	3 (FTP, Telnet, open DNS)	0	Service scan validation

Sustainability Plan

Upon completion of Phases 1 through 5, CRRWA will maintain its improved cybersecurity posture through:

- **Continuous monitoring**: Sentinel OT will provide ongoing passive monitoring and vulnerability alerting as an annual subscription ($9,600/year), funded through CRRWA's operating budget.
- **Quarterly review**: Sentinel OT will deliver quarterly posture reports identifying any new exposures, advisories affecting deployed equipment, or configuration drift.
- **Change management**: Phase 5 documentation includes updated access procedures and change management processes for plant operations staff.
- **Capital planning**: If the 1769-L18ER platform does not support CIP Security, controller replacement will be included in the FY2027 capital budget with a cost estimate provided during Phase 4.

Organizational Capability

Sentinel OT (Cybersecurity Contractor)

Sentinel OT is a cybersecurity intelligence firm specializing in operational technology and industrial control system security for critical infrastructure operators. Sentinel OT's capabilities include:

- Passive reconnaissance and vulnerability intelligence for OT/ICS environments
- On-site assessment of SCADA, DCS, and PLC-based control systems
- Network architecture design per IEC 62443 zone and conduit models
- Compliance documentation for AWIA, NIST SP 800-82, and CISA CPG frameworks

- Grant application support for SLCGP, DWSRF, and WIFIA-funded cybersecurity projects

CRRWA (Applicant)

Cedar Ridge Regional Water Authority operates one surface water treatment plant and a distribution system serving approximately 14,200 residents. CRRWA employs three full-time operators and contracts with a systems integrator for SCADA maintenance. CRRWA has not previously conducted a formal cybersecurity assessment of its OT infrastructure, which underscores the need for this engagement.

Supplementary Documentation

The following companion documents are available upon request:

1. **Vulnerability Intelligence Report** (Report ID: pbcwud_cedar-ridge_sample_2026-04-15). Detailed findings from the April 2026 passive intelligence assessment.
2. **Project Scope and Statement of Work**. Task-level descriptions, acceptance criteria, assumptions, and exclusions for all six project phases.
3. **CRRWA Cybersecurity Plan**. Alignment to the 16 required elements of the state cybersecurity plan (to be completed as part of the application process).

Sample document. Cedar Ridge Regional Water Authority is a fictional operator used to illustrate the format and depth of a Sentinel OT grant application narrative. No real utility, facility, or IP address is depicted. This document accompanies the sample Vulnerability Intelligence Report and Project Scope (Report ID: pbcwud_cedar-ridge_sample_2026-04-15).